



Data Analytics &
Data Engineering

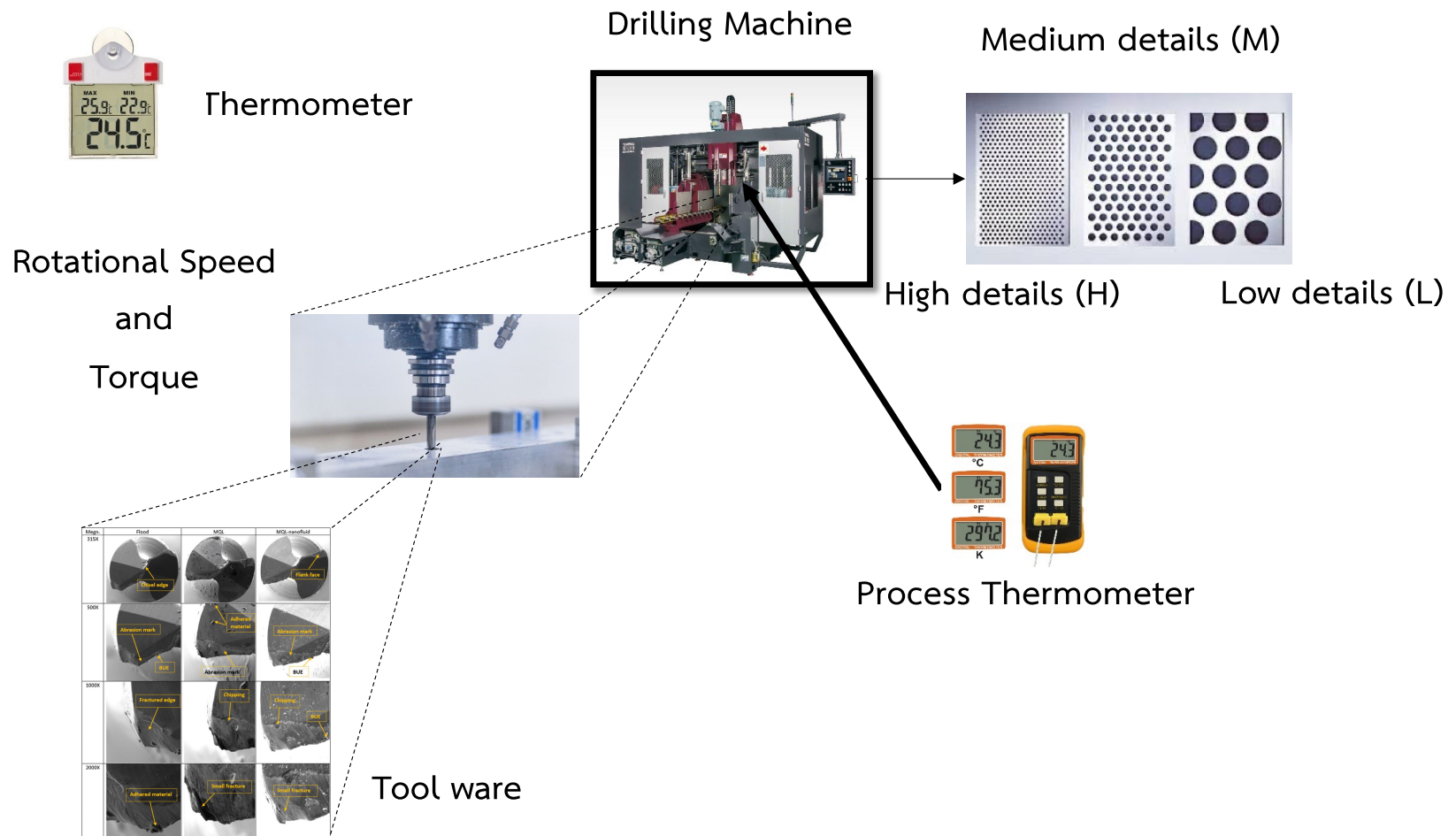
KNIME Case Study:
Classification Model

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Waranyu Wongseree

สมมุติฐานของกรณีศึกษาการจำแนก (Assumptions of Classification Case Study)



Case Study: Predictive maintenance



Original dataset

Possible Features

Possible Class

UDI	Product ID	Type	Air temperature [K]	Process temperature [K]	Rotational speed [rpm]	Torque [Nm]	Tool wear [min]	Machine failure	TWF	HDF	PWF	OSF	RNF
1	M14860	M	298.1	308.6	1551	42.8	0	0	0	0	0	0	0
2	L47181	L	298.2	308.7	1408	46.3	3	0	0	0	0	0	0
3	L47182	L	298.1	308.5	1498	49.4	5	0	0	0	0	0	0
4	L47183	L	298.2	308.6	1433	39.5	7	0	0	0	0	0	0
5	L47184	L	298.2	308.7	1408	40	9	0	0	0	0	0	0
6	M14865	M	298.1	308.6	1425	41.9	11	0	0	0	0	0	0
7	L47186	L	298.1	308.6	1558	42.4	14	0	0	0	0	0	0
8	L47187	L	298.1	308.6	1527	40.2	16	0	0	0	0	0	0
9	M14868	M	298.3	308.7	1667	28.6	18	0	0	0	0	0	0
10	M14869	M	298.5	309	1741	28	21	0	0	0	0	0	0
11	H29424	H	298.4	308.9	1782	23.9	24	0	0	0	0	0	0
12	H29425	H	298.6	309.1	1423	44.3	29	0	0	0	0	0	0
13	M14872	M	298.6	309.1	1339	51.1	34	0	0	0	0	0	0
14	M14873	M	298.6	309.2	1742	30	37	0	0	0	0	0	0
15	L47194	L	298.6	309.2	2035	19.6	40	0	0	0	0	0	0
16	L47195	L	298.6	309.2	1542	48.4	42	0	0	0	0	0	0
17	M14876	M	298.6	309.2	1311	46.6	44	0	0	0	0	0	0
18	M14877	M	298.7	309.2	1410	45.6	47	0	0	0	0	0	0
19	H29432	H	298.8	309.2	1306	54.5	50	0	0	0	0	0	0
20	M14879	M	298.9	309.3	1632	32.5	55	0	0	0	0	0	0
21	H29434	H	298.9	309.3	1375	42.7	58	0	0	0	0	0	0
22	L47201	L	298.8	309.3	1450	44.8	63	0	0	0	0	0	0
23	M14882	M	298.9	309.3	1581	30.7	65	0	0	0	0	0	0
24	L47203	L	299	309.4	1758	25.7	68	0	0	0	0	0	0
25	M14884	M	299	309.4	1561	37.3	70	0	0	0	0	0	0

Assumptions

Selected Features

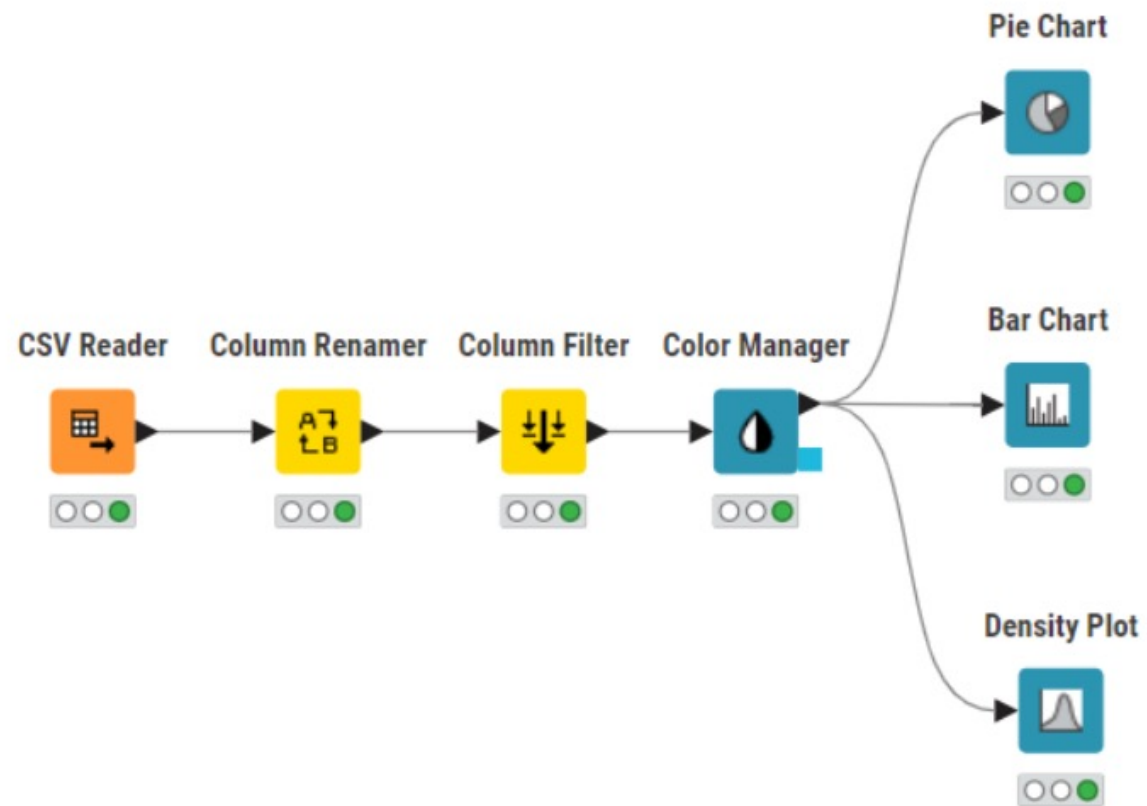
Selected Class

Type	Air temperature [K]	Process temperature [K]	Rotational speed [rpm]	Torque [Nm]	Tool wear [min]	Machine failure
M	298.1	308.6	1551	42.8	0	0
L	298.2	308.7	1408	46.3	3	0
L	298.1	308.5	1498	49.4	5	0
L	298.2	308.6	1433	39.5	7	0
L	298.2	308.7	1408	40	9	0
M	298.1	308.6	1425	41.9	11	0
L	298.1	308.6	1558	42.4	14	0
L	298.1	308.6	1527	40.2	16	0
M	298.3	308.7	1667	28.6	18	0
M	298.5	309	1741	28	21	0
H	298.4	308.9	1782	23.9	24	0
H	298.6	309.1	1423	44.3	29	0
M	298.6	309.1	1339	51.1	34	0
M	298.6	309.2	1742	30	37	0
L	298.6	309.2	2035	19.6	40	0
L	298.6	309.2	1542	48.4	42	0
M	298.6	309.2	1311	46.6	44	0
M	298.7	309.2	1410	45.6	47	0
H	298.8	309.2	1306	54.5	50	0
M	298.9	309.3	1632	32.5	55	0
H	298.9	309.3	1375	42.7	58	0
L	298.8	309.3	1450	44.8	63	0
M	298.9	309.3	1581	30.7	65	0
L	299	309.4	1758	25.7	68	0
M	299	309.4	1561	37.3	70	0

KNIME Visualization

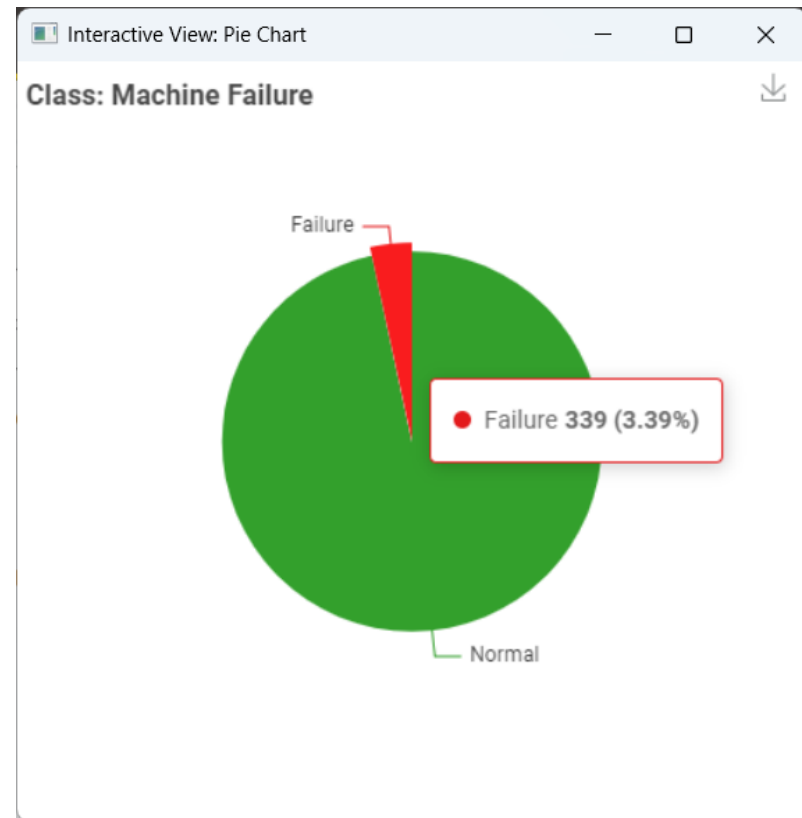


KNIME: Univariate Visualization



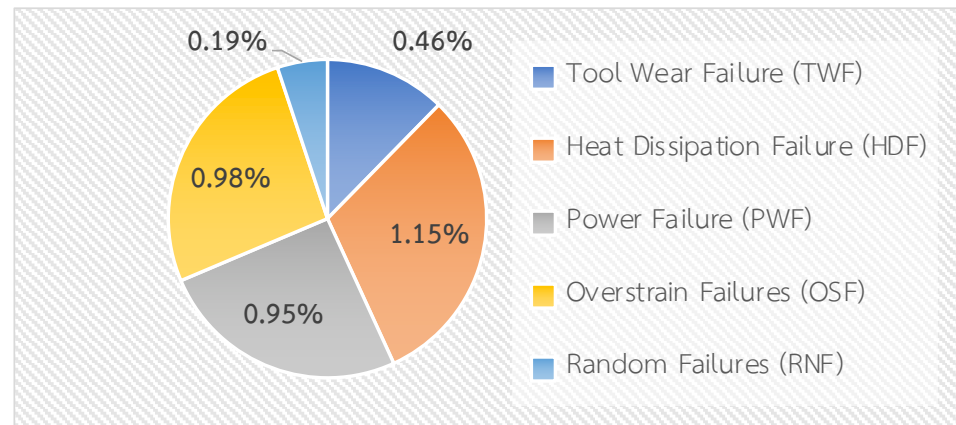
Class: Normal / Failure

- Normal (96.6%)
- Failure (3.4%)
 - Tool Wear Failure (TWF)
 - Heat Dissipation Failure (HDF)
 - Power Failure (PWF)
 - Overstrain Failures (OSF)
 - Random Failures (RNF)



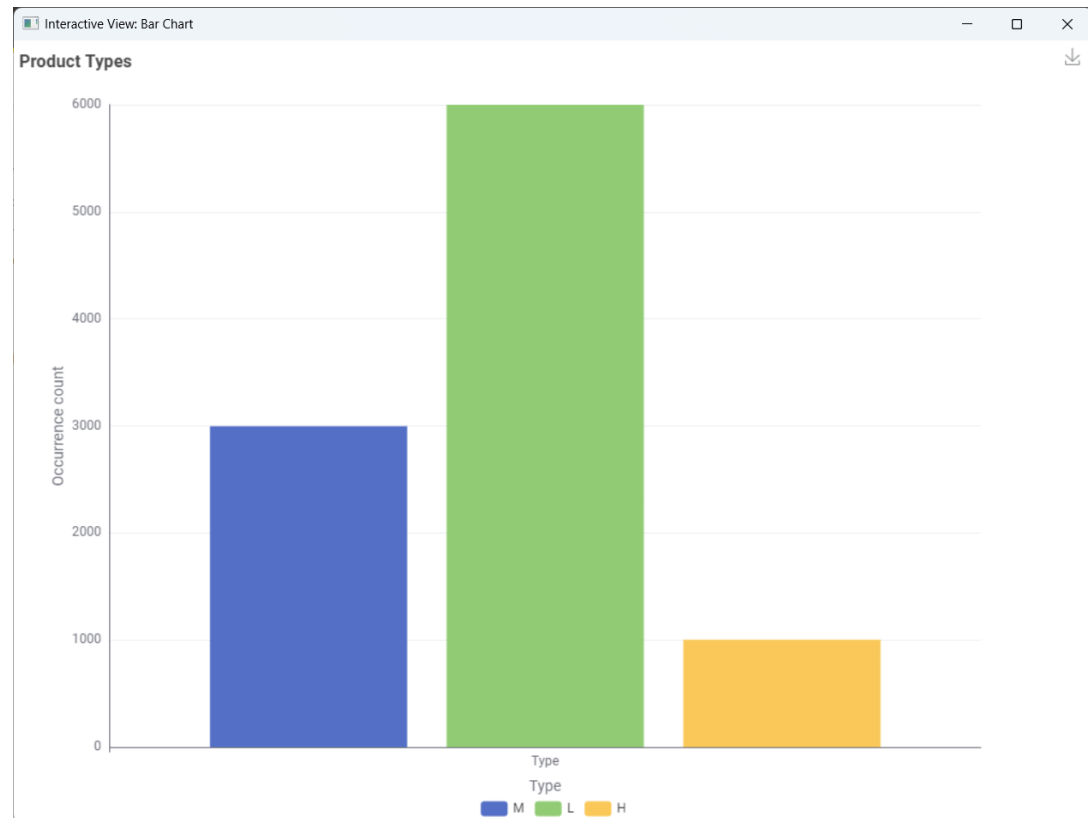
Class: Failure

- Tool Wear Failure (TWF)
- Heat Dissipation Failure (HDF)
- Power Failure (PWF)
- Overstrain Failures (OSF)
- Random Failures (RNF)



ตัวแปรที่ 1: ชนิดของสินค้า (Product Type)

- Low (60%)
- Medium (30%)
- High (10%)



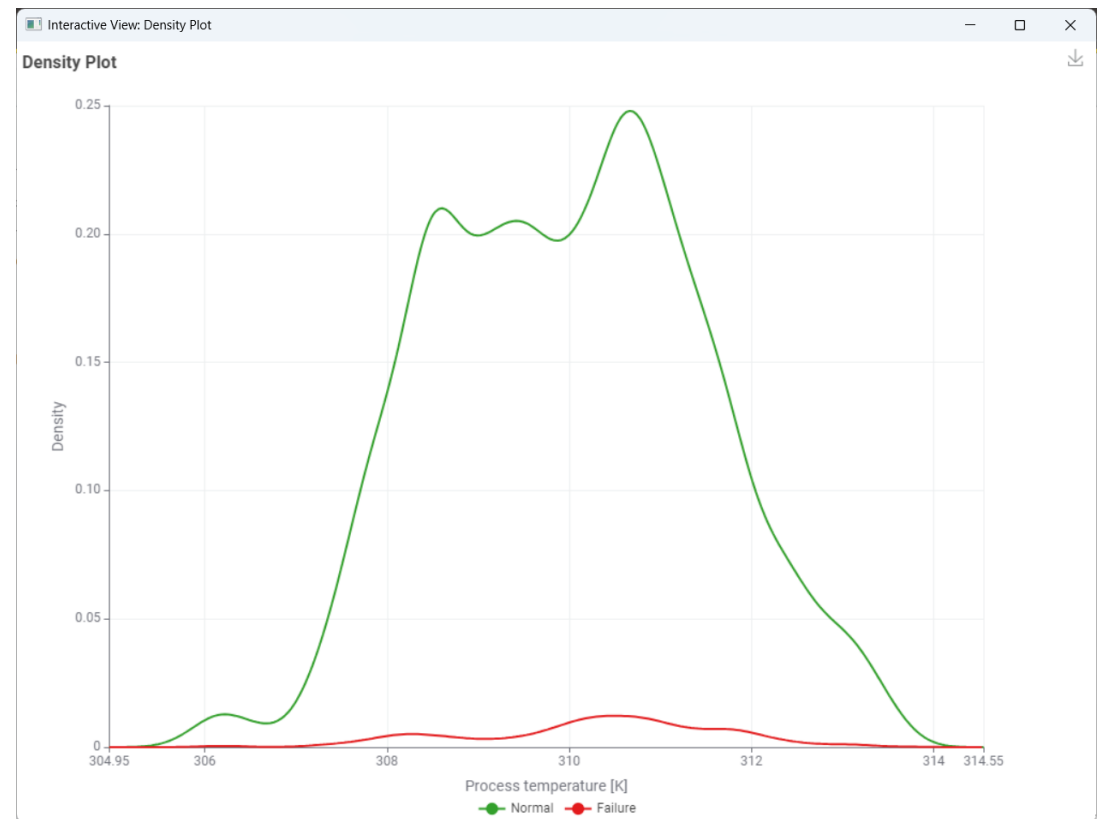
ตัวแปรที่ 2: Air Temperature

- Numeric
- หน่วย: เคลวิน (Kelvin)



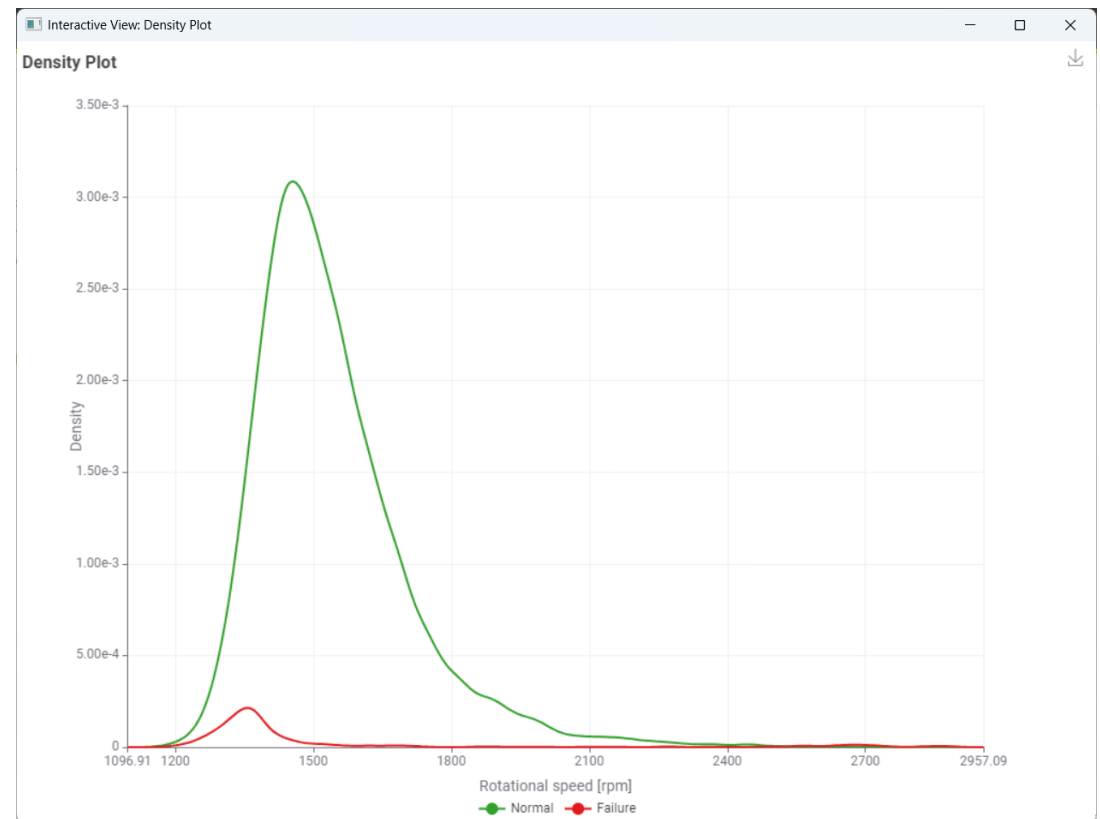
ตัวแปรที่ 3: Process Temperature

- Numeric
- หน่วย: เคลวิน (Kelvin)



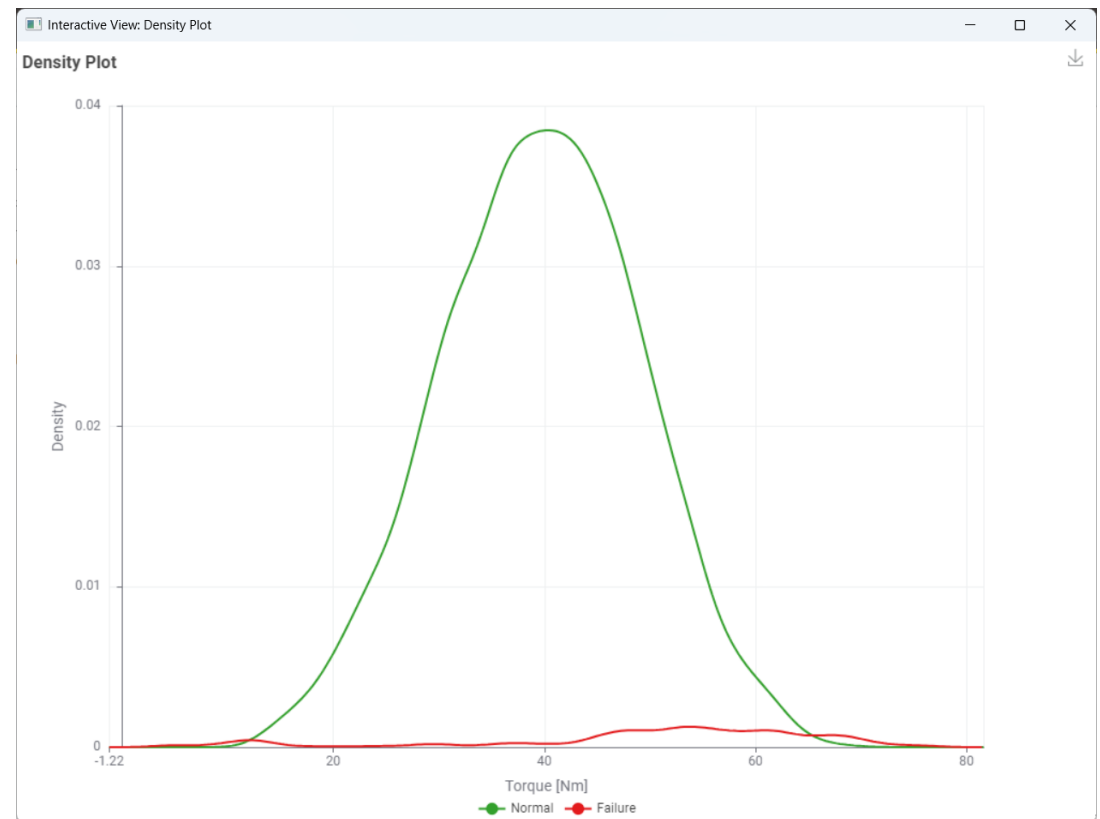
ตัวแปรที่ 4: Rotational Speed

- มอเตอร์ขนาด 2860 วัตต์
- Numeric
- หน่วย: รอบต่อนาที



ตัวแปรที่ 5: แรงบิด (Torque)

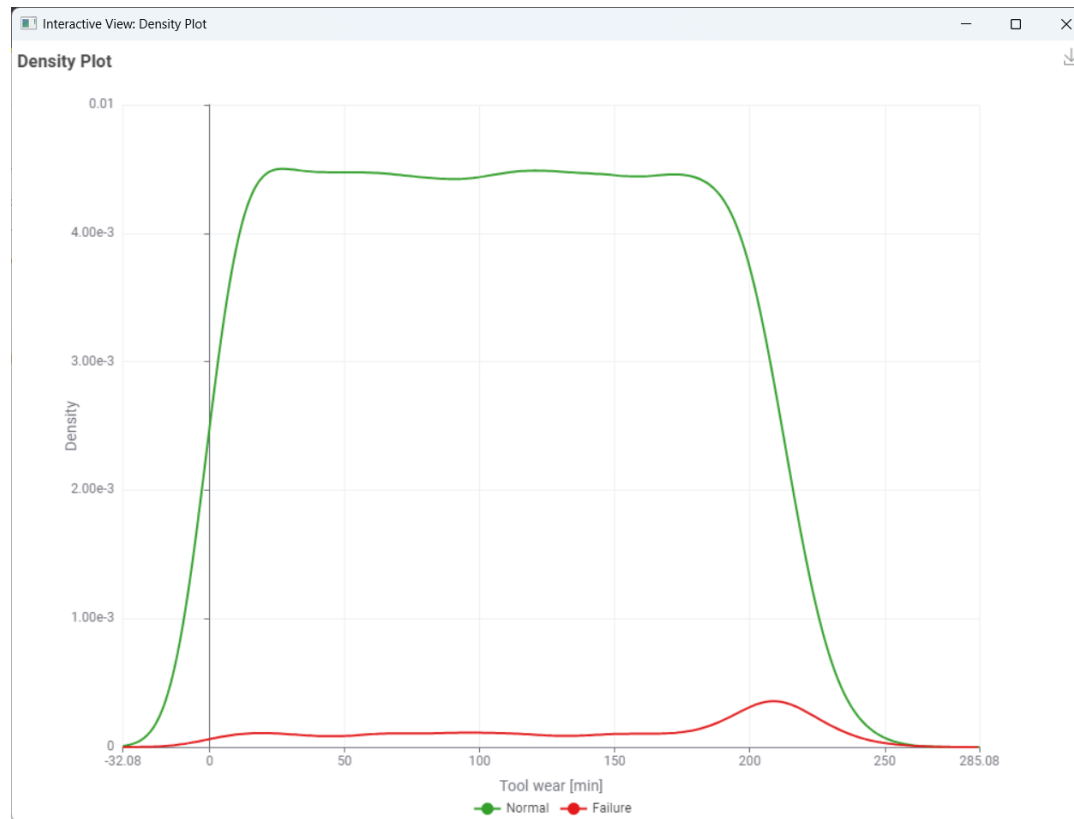
- Numeric
- หน่วย: นิวตันเมตร



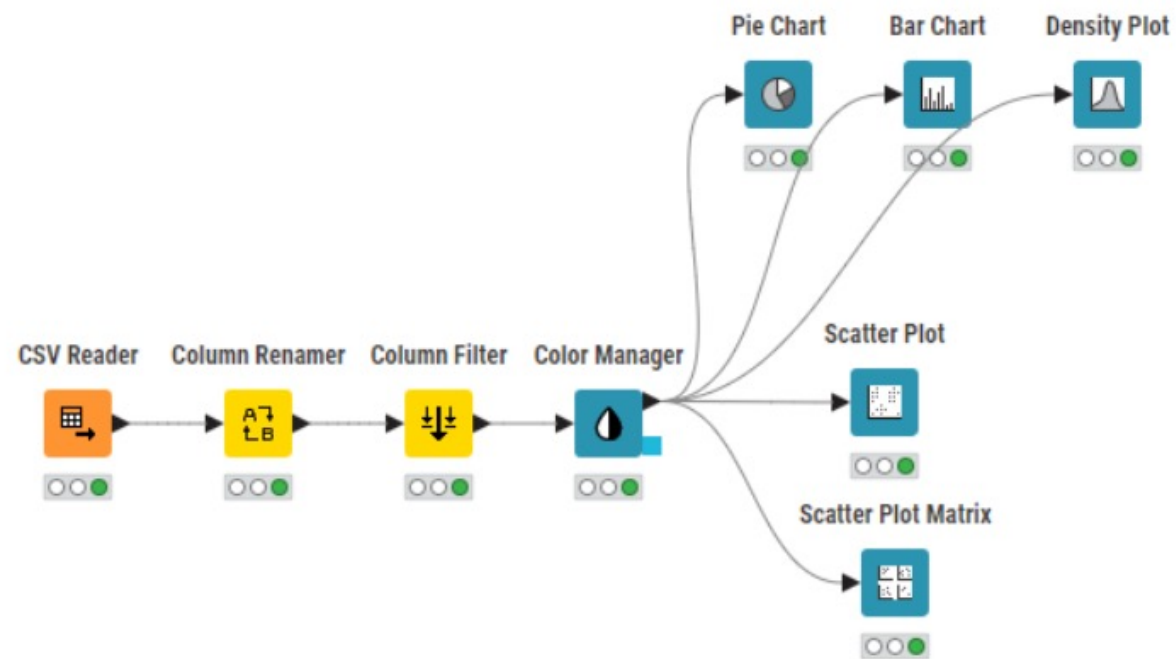
ตัวแปรที่ 6: การสึกหรอของมีดตัด (Tool Wear)

- Numeric
- หน่วย: นาที
 - การผลิตสินค้าที่มีความละเอียดต่ำ (Low, L) การสึกหรอของมีดตัดเป็น 2 นาที
 - การผลิตสินค้าที่มีความละเอียดปานกลาง (Medium, M) การสึกหรอของมีดตัดเป็น 3 นาที
 - การผลิตสินค้าที่มีความละเอียดสูง (High, H) การสึกหรอของมีดตัดเป็น 5 นาที

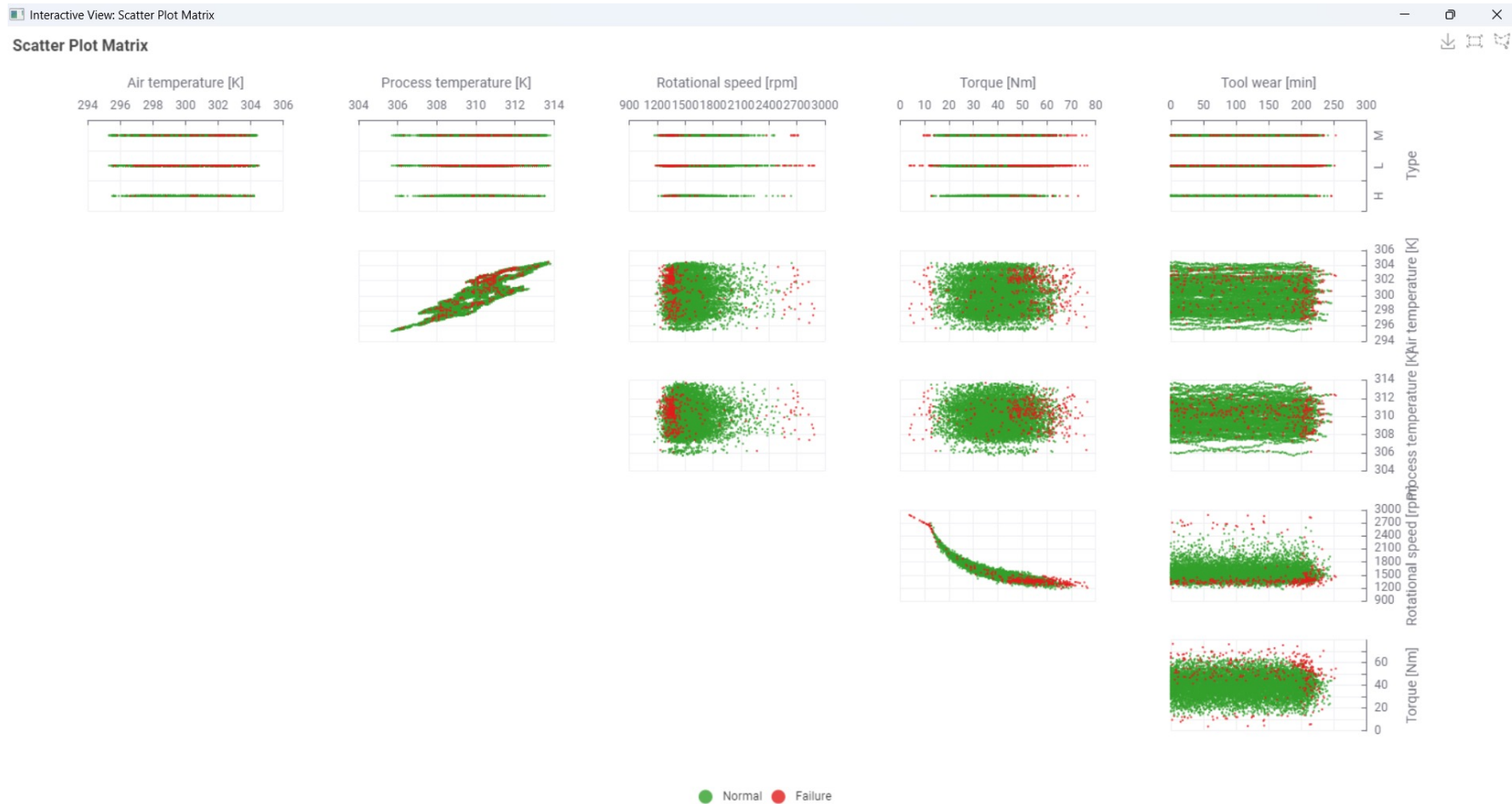
ตัวแปรที่ 6: การสึกหรอของมีดตัด (Tool Wear)



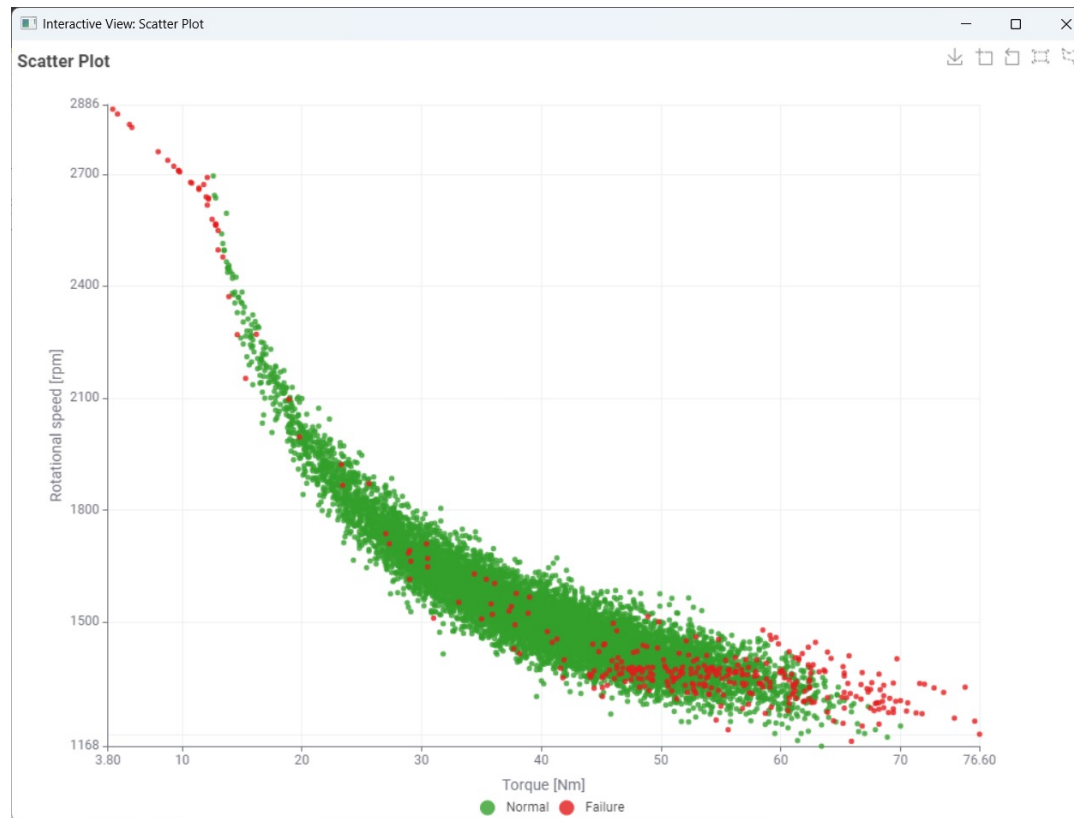
KNIME: Multivariate Visualization



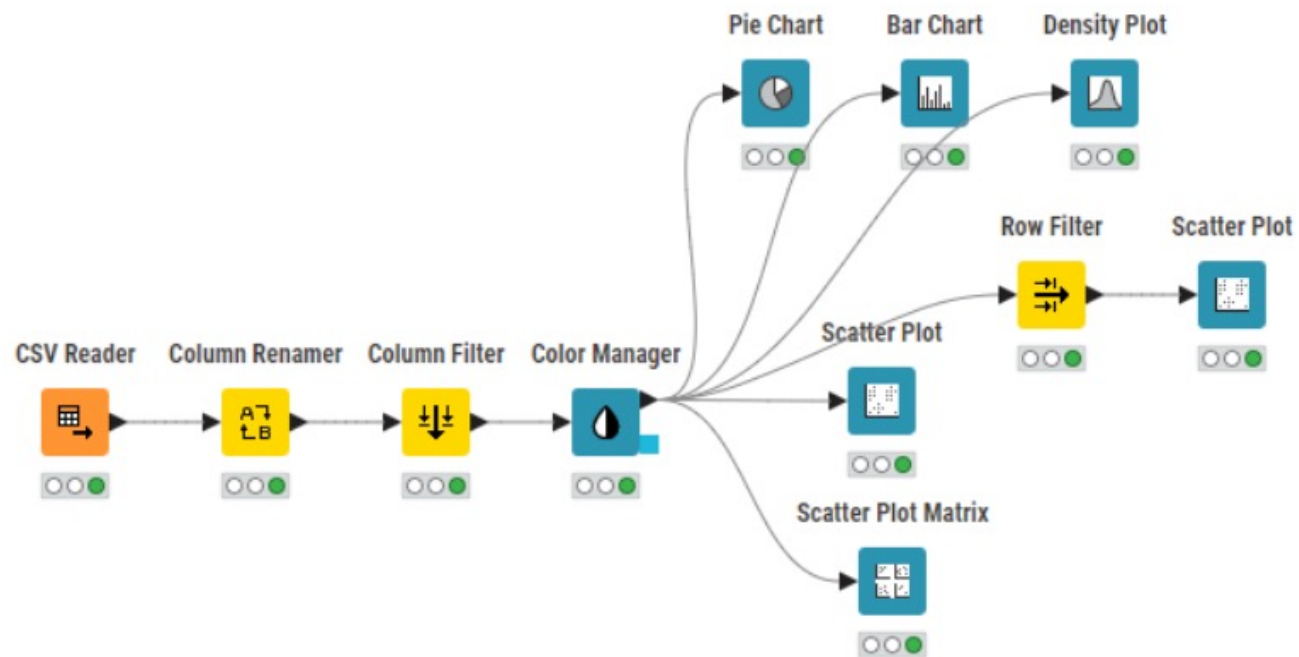
KNIME: Scatter Plot Matrix



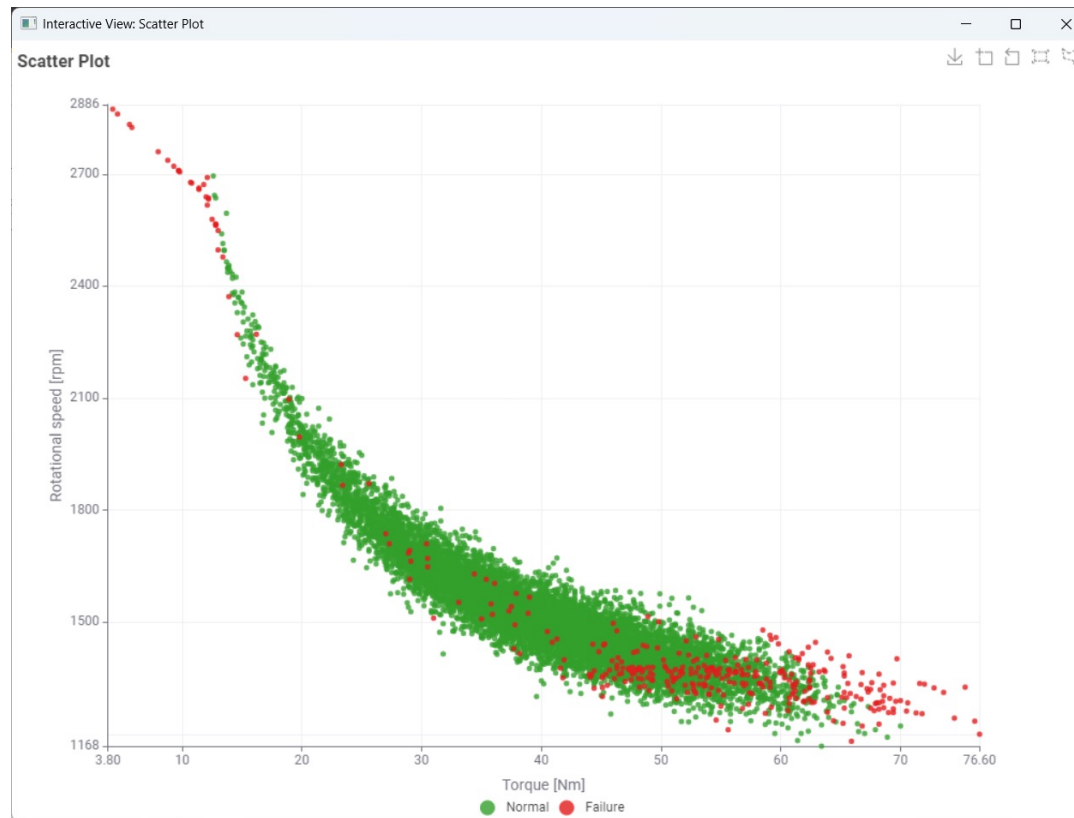
KNIME: Scatter Plot



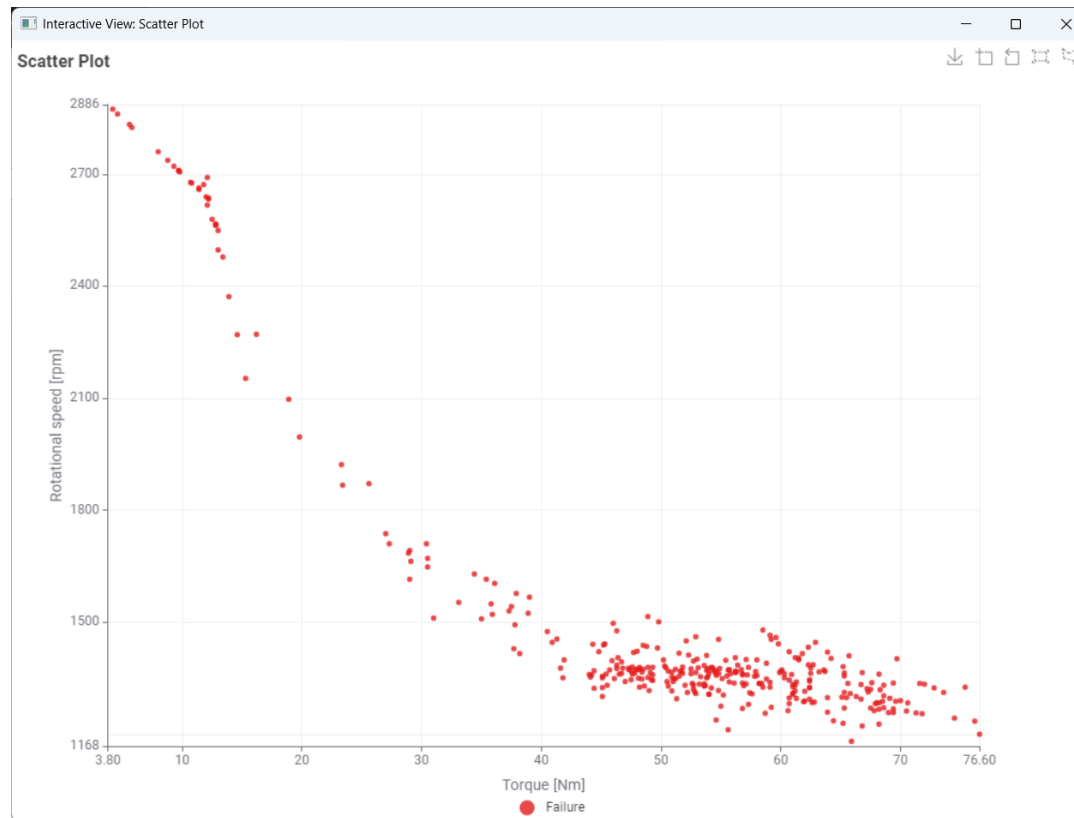
KNIME: Targeted Scatter Plot



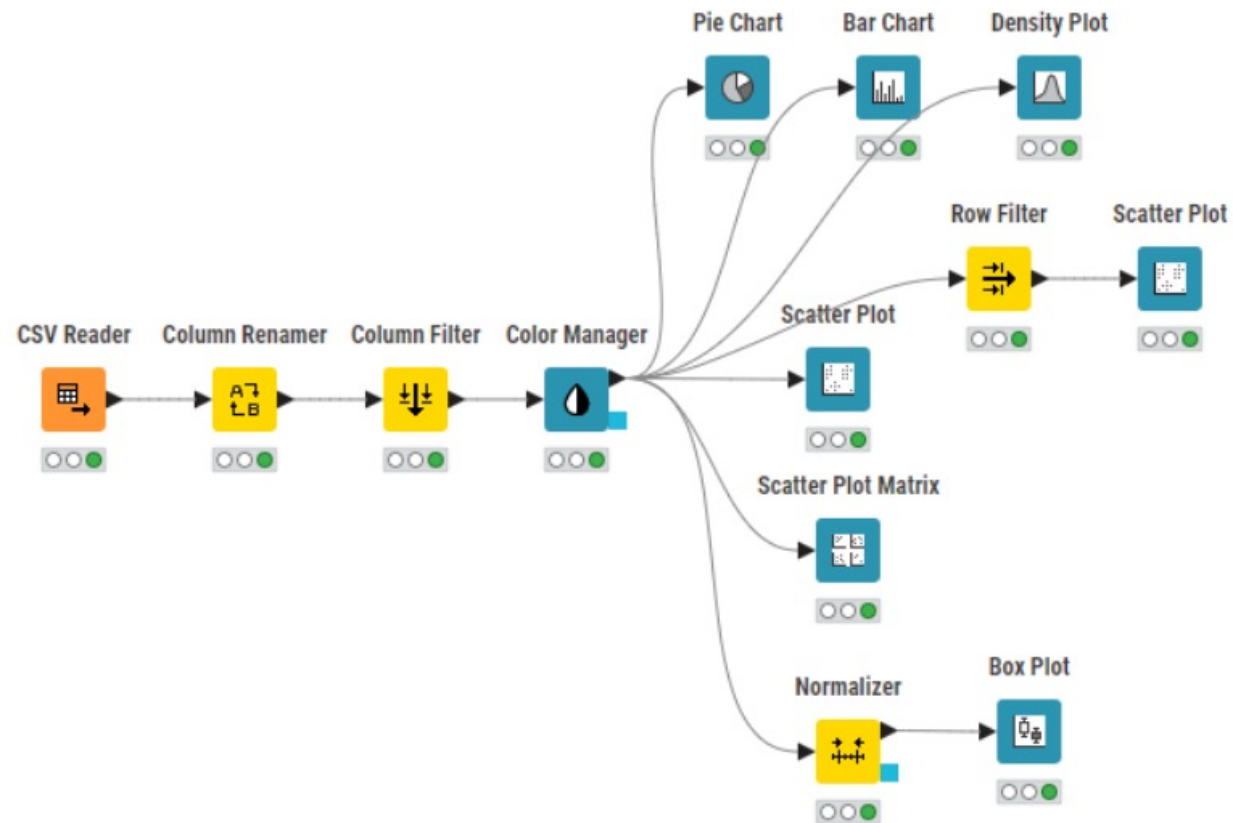
KNIME: Scatter Plot



KNIME: Targeted Failure Labels

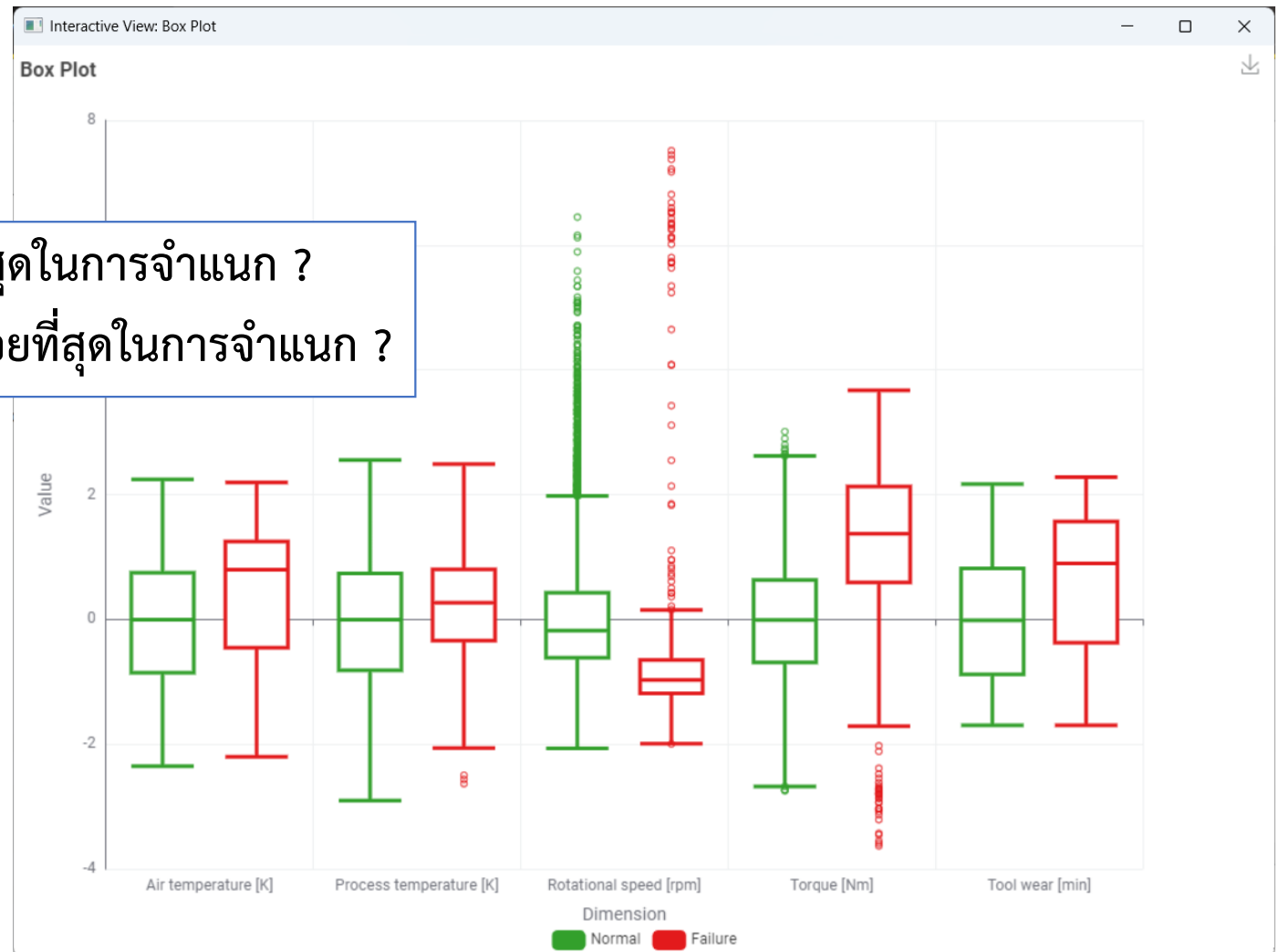


KNIME: Multivariate Visualization

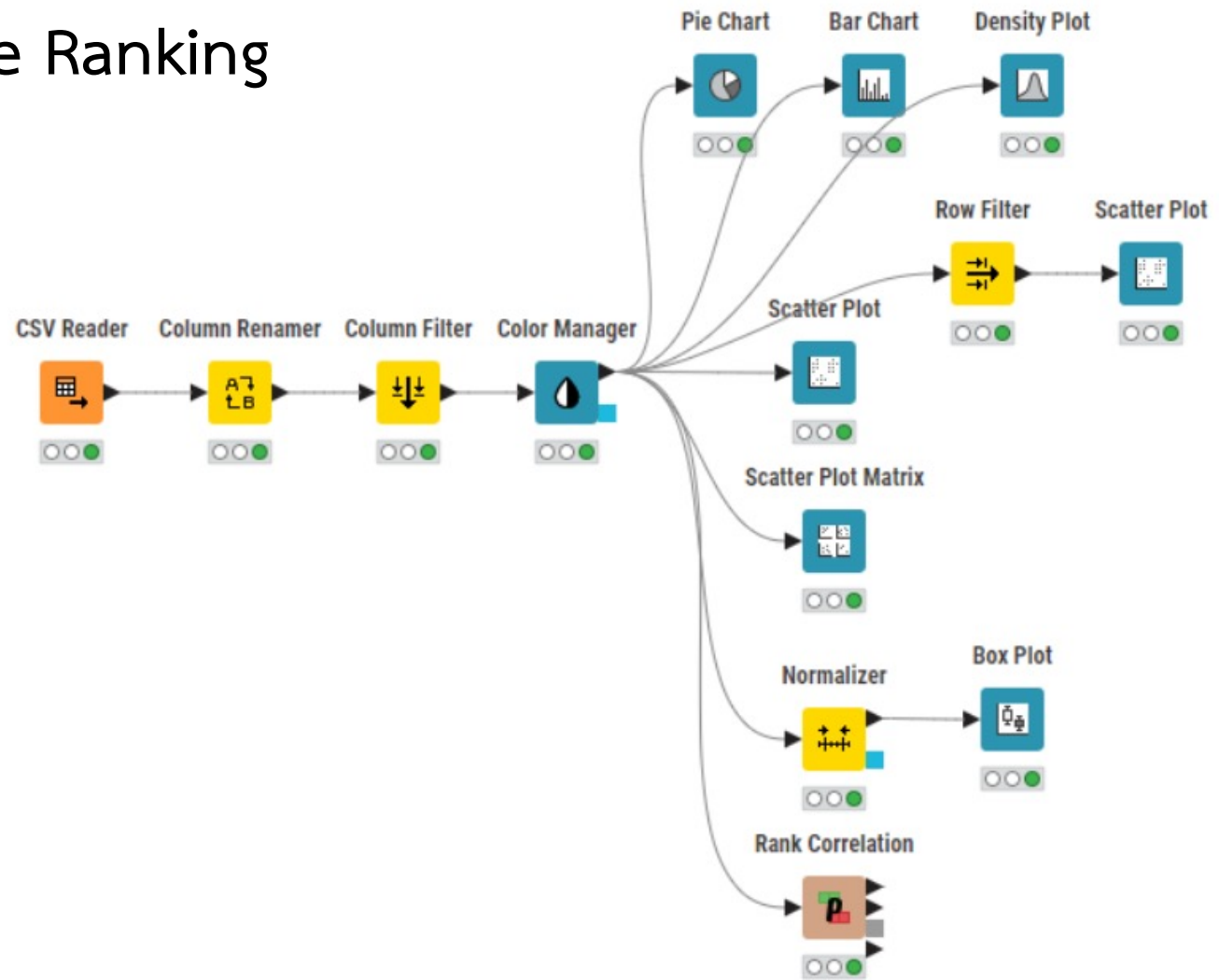


KNIME: Box Plot

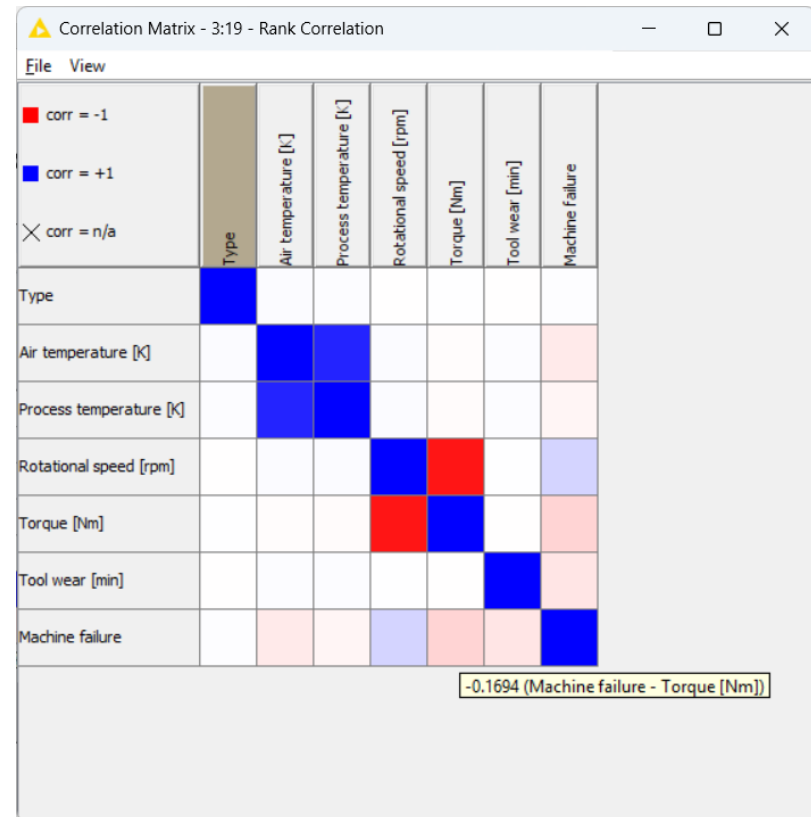
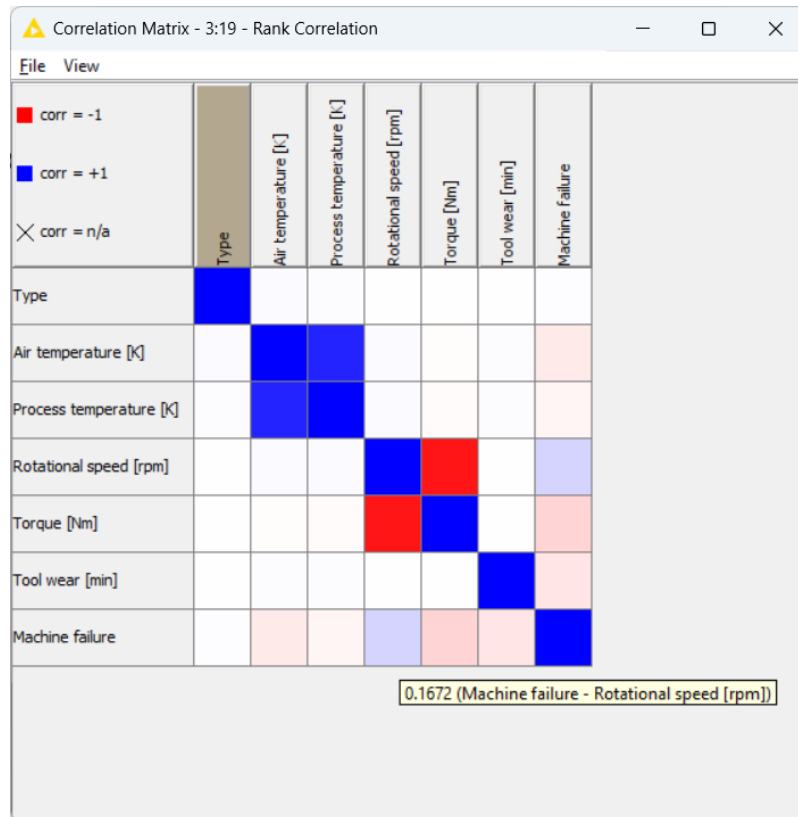
- 1) ตัวแปรใดสำคัญที่สุดในการจำแนก ?
- 2) ตัวแปรใดสำคัญน้อยที่สุดในการจำแนก ?



KNIME: Feature Ranking



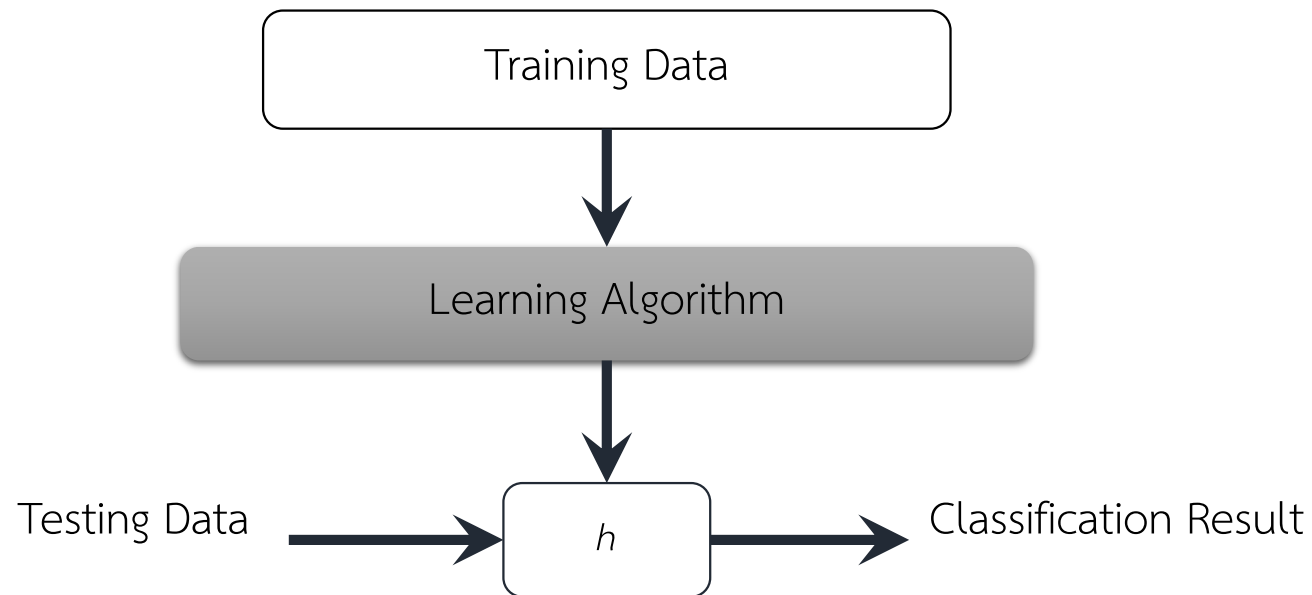
KNIME: Box Plot



KNIME: Modeling and Evaluation



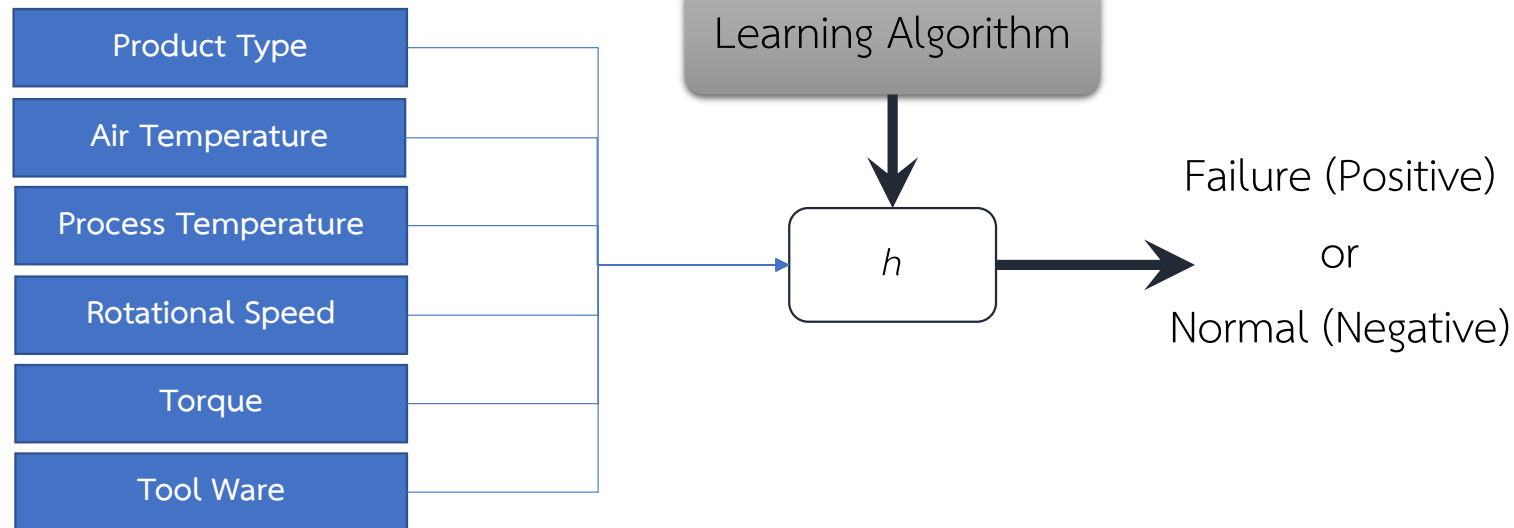
Problem Formulation



Imbalanced Binary Classification Modeling

10,000 samples, 6 features
Extreme Imbalanced Binary Classes:
- Tiny positive class (Failure $\approx$ 3%)

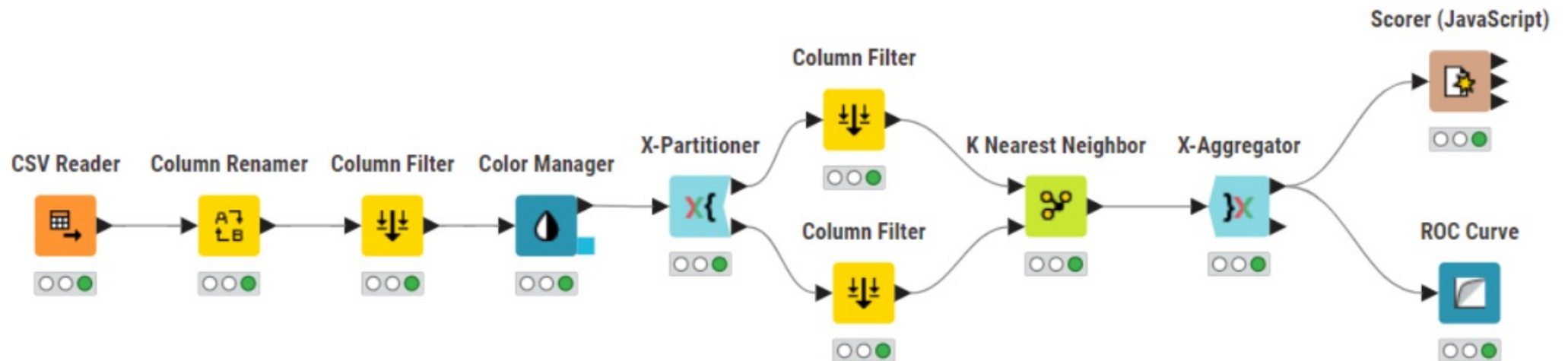
Type	Air temperature [K]	Process temperature [K]	Rotational speed [rpm]	Torque [Nm]	Tool wear [min]	Machine failure
M	298.1	308.6	1551	42.8	0	0
L	298.2	308.7	1408	46.3	3	0
L	298.1	308.5	1498	49.4	5	0
L	298.2	308.6	1433	39.5	7	0
L	298.2	308.7	1408	40	9	0
M	298.1	308.6	1425	41.9	11	0
L	298.1	308.6	1558	42.4	14	0
L	298.1	308.6	1527	40.2	16	0
M	298.3	308.7	1667	28.6	18	0
M	298.5	309	1741	28	21	0
H	298.4	308.9	1782	23.9	24	0
H	298.6	309.1	1423	44.3	29	0
M	298.6	309.1	1339	51.1	34	0
M	298.6	309.2	1742	30	37	0
L	298.6	309.2	2035	19.6	40	0
L	298.6	309.2	1542	48.4	42	0
M	298.6	309.2	1311	46.6	44	0
M	298.7	309.2	1410	65.6	47	0
H	298.8	309.2	1306	54.5	50	0
M	298.9	309.3	1632	32.5	55	0
H	298.9	309.3	1375	42.7	58	0
L	298.8	309.3	1450	44.8	63	0
M	298.9	309.3	1581	30.7	65	0
L	299	309.4	1758	25.7	68	0
M	299	309.4	1561	37.3	70	0



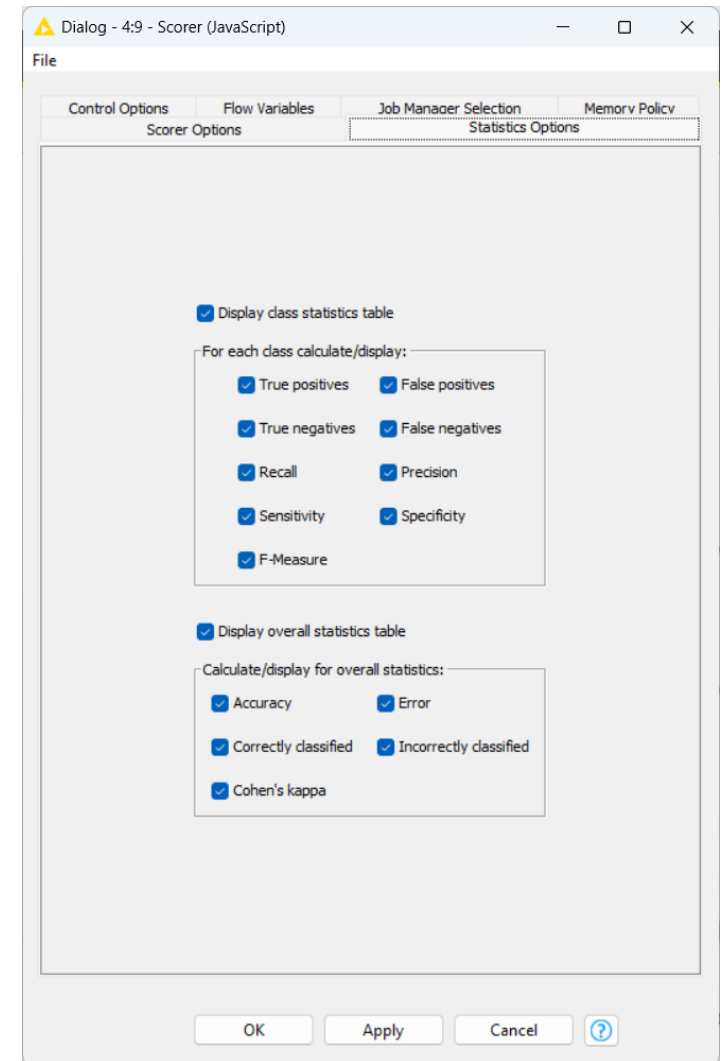
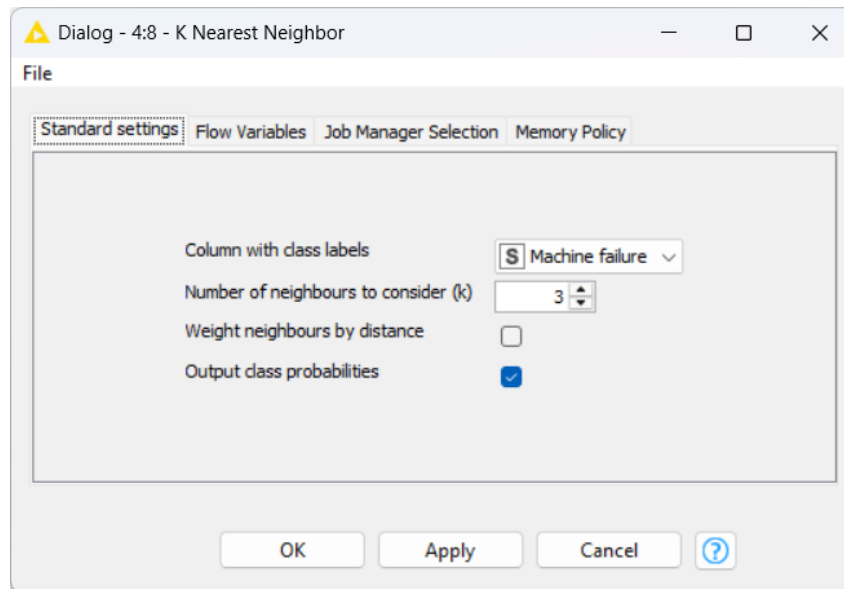
KNIME: K-Nearest Neighbor



KNIME: K-Nearest Neighbor [10CV]



Important Setting: K-Nearest Neighbor



K-Nearest Neighbor: Classification Metrics

Confusion Matrix

Scorer View

Confusion Matrix

	Failure (Predicted)	Normal (Predicted)	
Failure (Actual)	84	255	24.78%
Normal (Actual)	69	9592	99.29%
	54.90%	97.41%	

Class Statistics

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	84	69	9592	255	24.78%	54.90%	24.78%	99.29%	34.15%
Normal	9592	255	84	69	99.29%	97.41%	99.29%	24.78%	98.34%

Overall Statistics

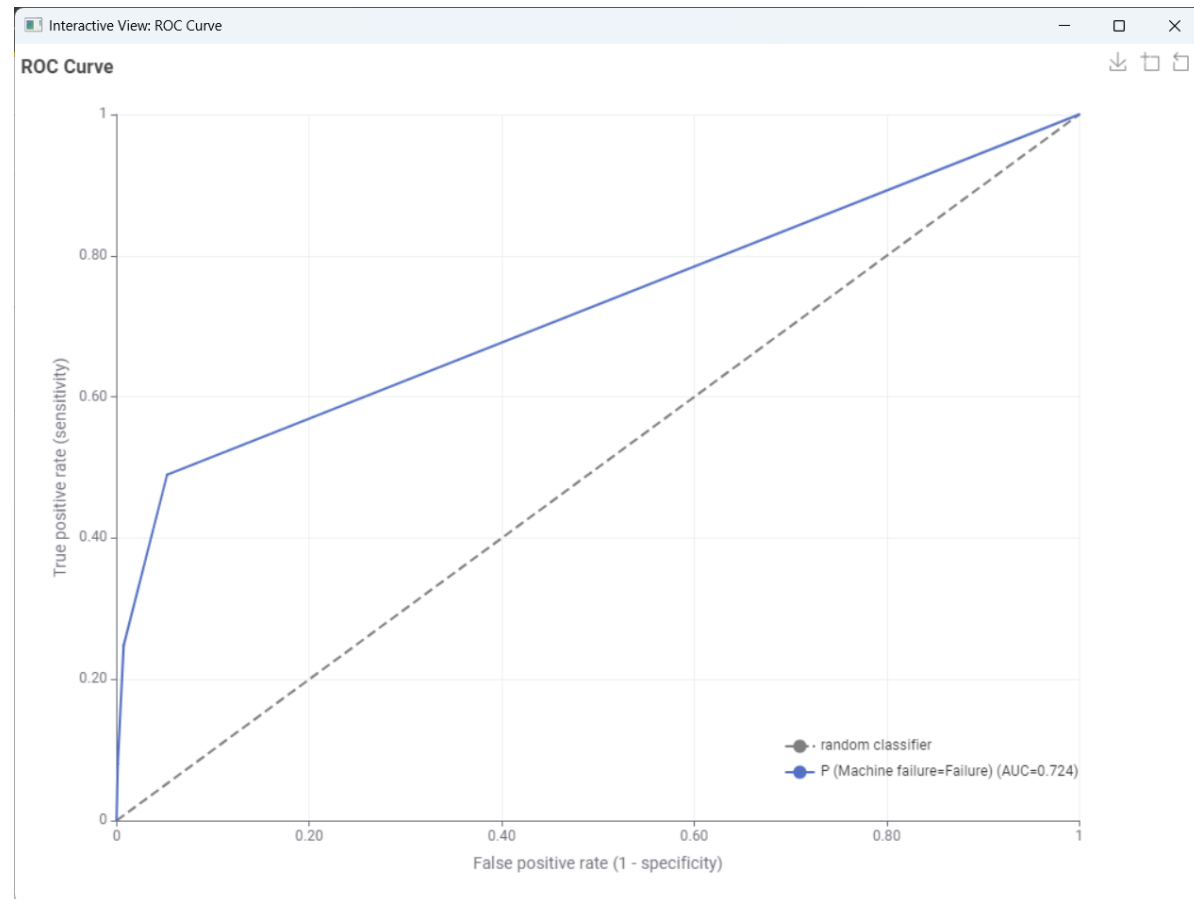
Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
96.76%	3.24%	0.327	9676	324

Reset

Apply

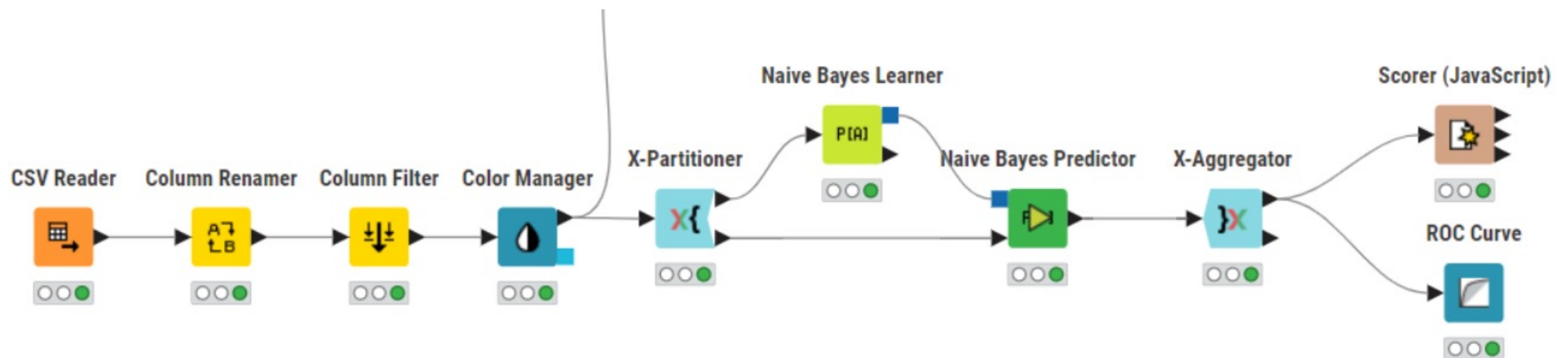
Close

K-Nearest Neighbor: ROC

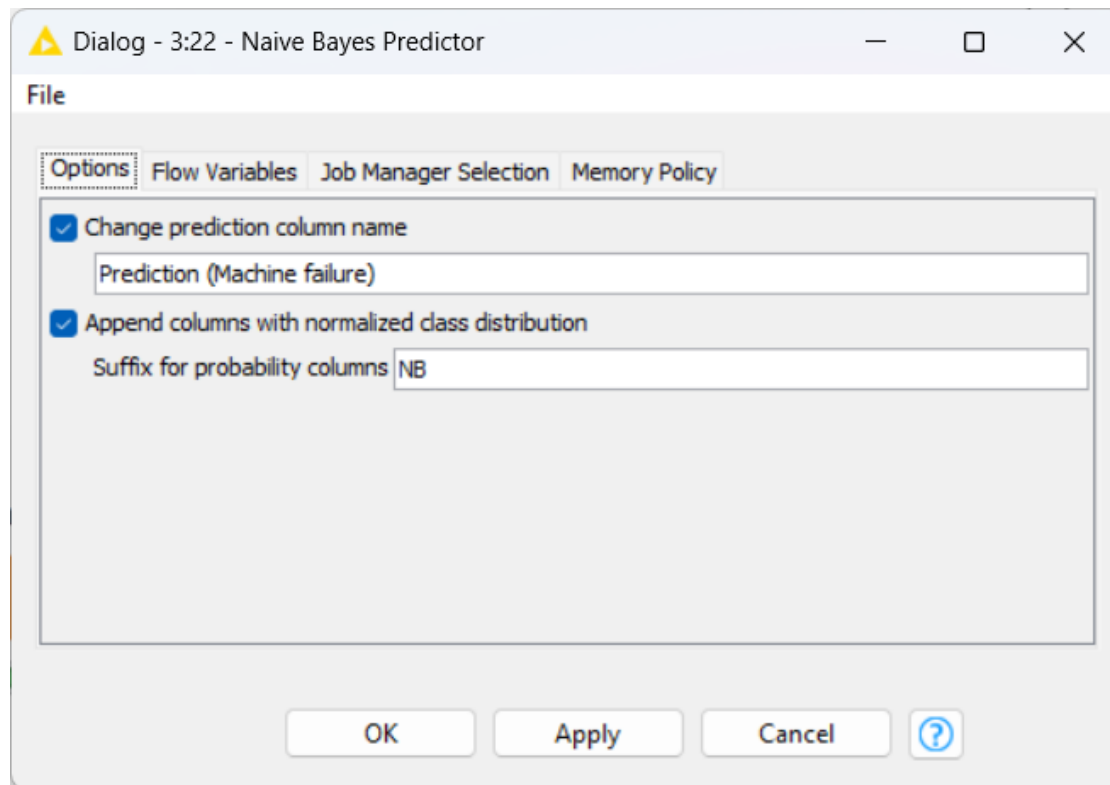


KNIME: Naïve Bayes

KNIME: Naïve Bayes



Important Setting: Naïve Bayes



Naïve Bayes: Classification Metrics

Confusion Matrix

Scorer View

Confusion Matrix

	Failure (Predicted)	Normal (Predicted)	
Failure (Actual)	34	305	10.03%
Normal (Actual)	8	9653	99.92%
	80.95%	96.94%	

Class Statistics

Class	True Positives	False Positives	True Negatives	False Negatives	Recall	Precision	Sensitivity	Specificity	F-measure
Failure	34	8	9653	305	10.03%	80.95%	10.03%	99.92%	17.85%
Normal	9653	305	34	8	99.92%	96.94%	99.92%	10.03%	98.40%

Overall Statistics

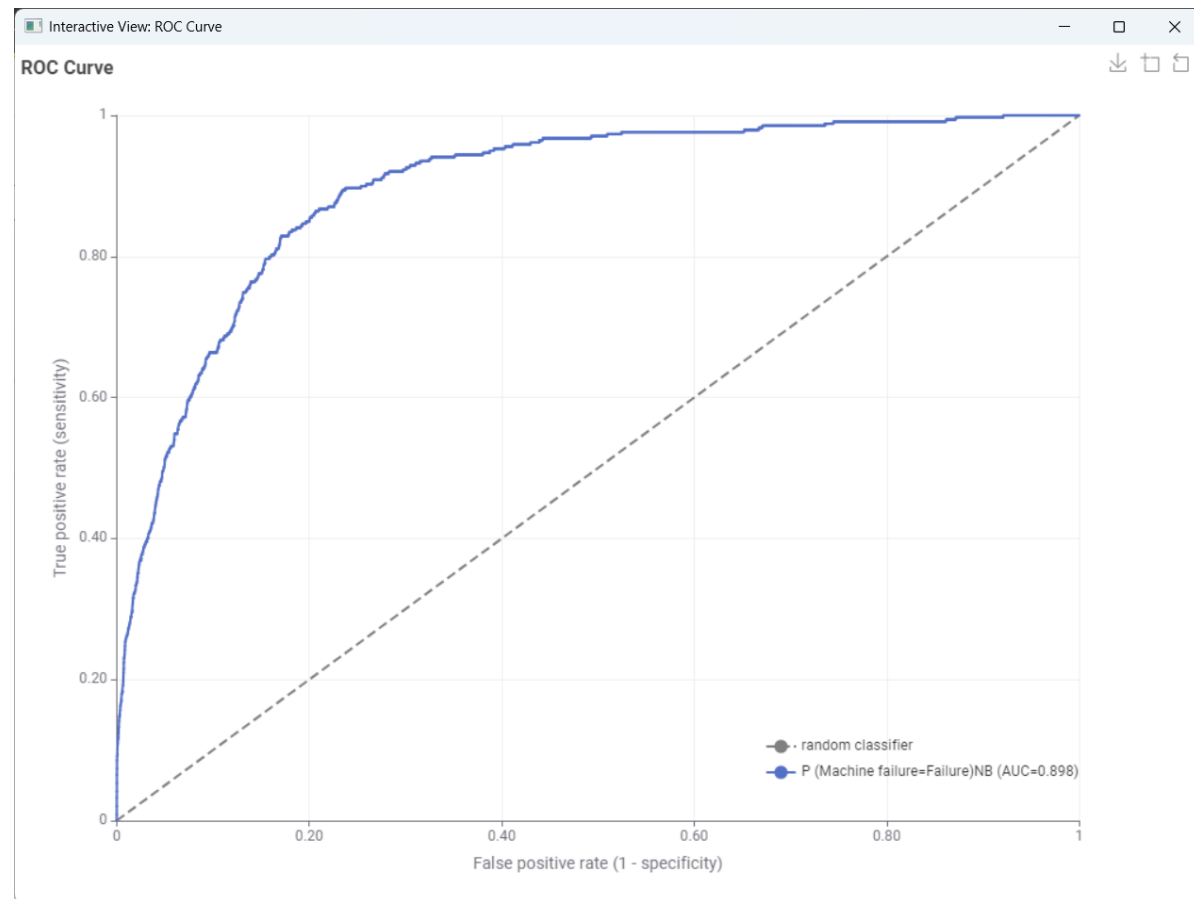
Overall Accuracy	Overall Error	Cohen's kappa (κ)	Correctly Classified	Incorrectly Classified
96.87%	3.13%	0.172	9687	313

Reset

Apply

Close

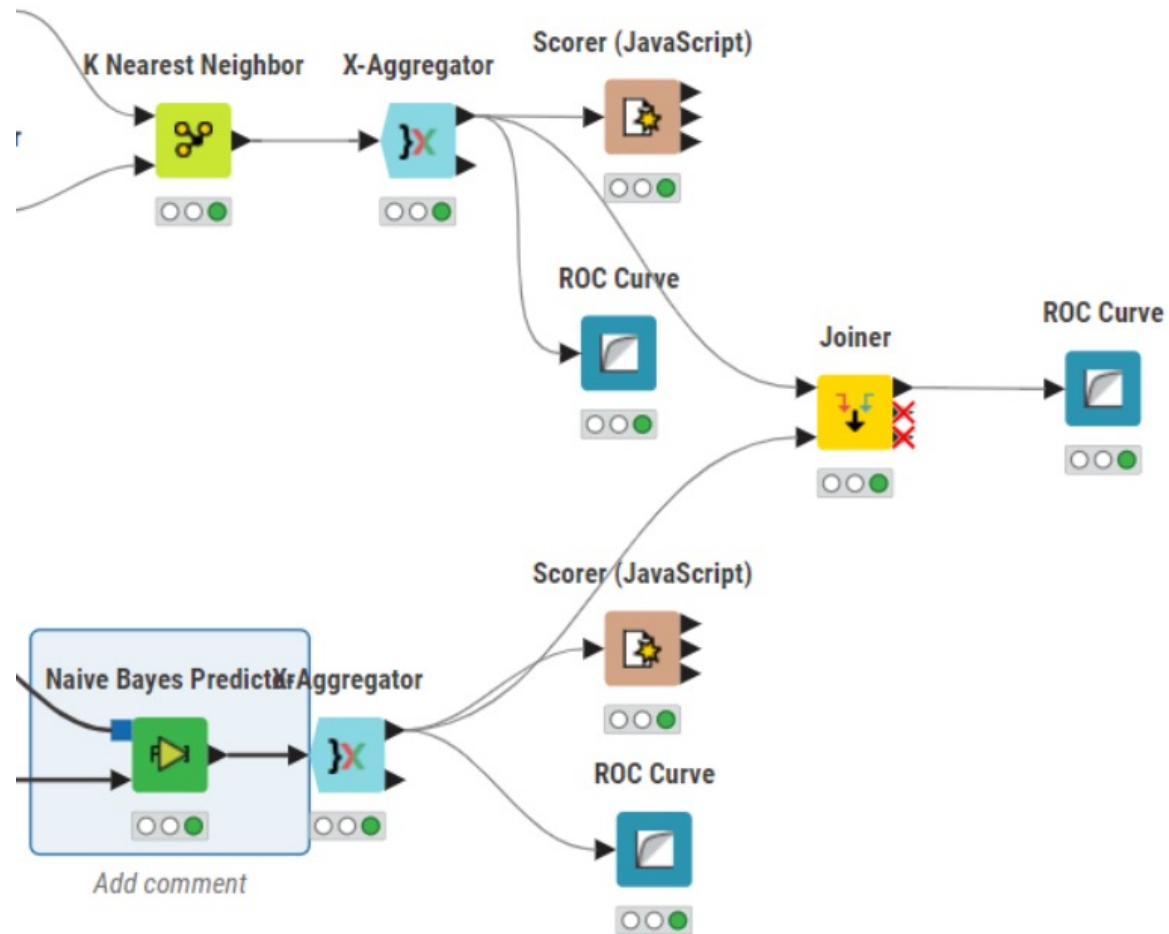
Naïve Bayes: ROC



KNIME: ROC Comparison



KNIME: ROC Comparison



Important Setting: Joiner

Dialog - 3:28 - Joiner

Include in Output

☐ Left unmatched rows

☐ Right unmatched rows

Output Columns

Top input ('left' table)

Manual Wildcard Regex

Search Aa

Excludes

- Air temperature [K]
- Process temperature [K]
- Rotational speed [rpm]
- Torque [Nm]
- Tool wear [min]
- Class [kNN]
- P (Machine failure=Normal)

Any unknown column

Includes

- Machine failure
- P (Machine failure=Failure)

Cancel Ok

Dialog - 3:28 - Joiner

Output Columns

Bottom input ('right' table)

Manual Wildcard Regex

Search Aa

Excludes

- Type
- Air temperature [K]
- Process temperature [K]
- Rotational speed [rpm]
- Torque [Nm]
- Tool wear [min]
- P (Machine failure=Normal)NB
- Prediction (Machine failure)

Any unknown column

Includes

- Machine failure
- P (Machine failure=Failure)NB

☐ Merge join columns

If there are duplicate column names

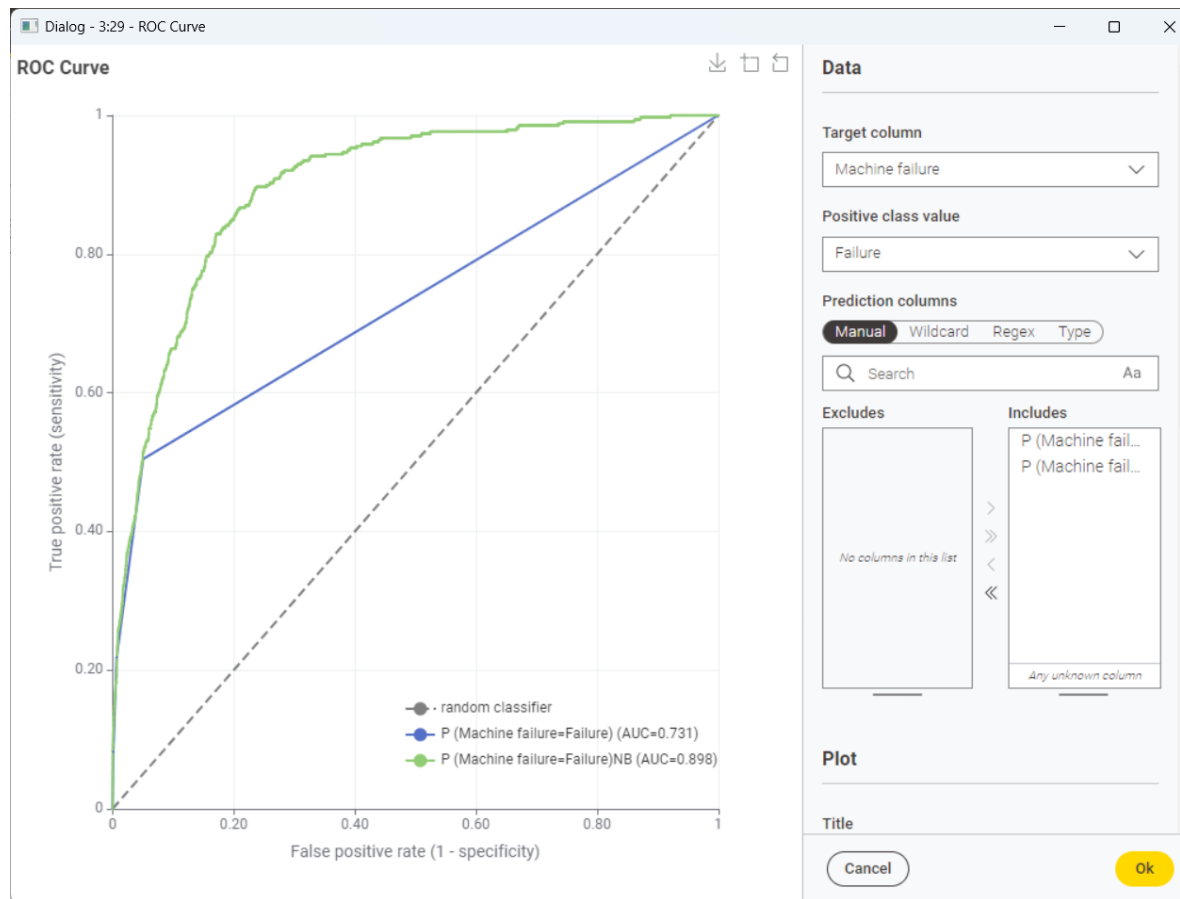
☒ Append custom suffix

☐ Do not execute

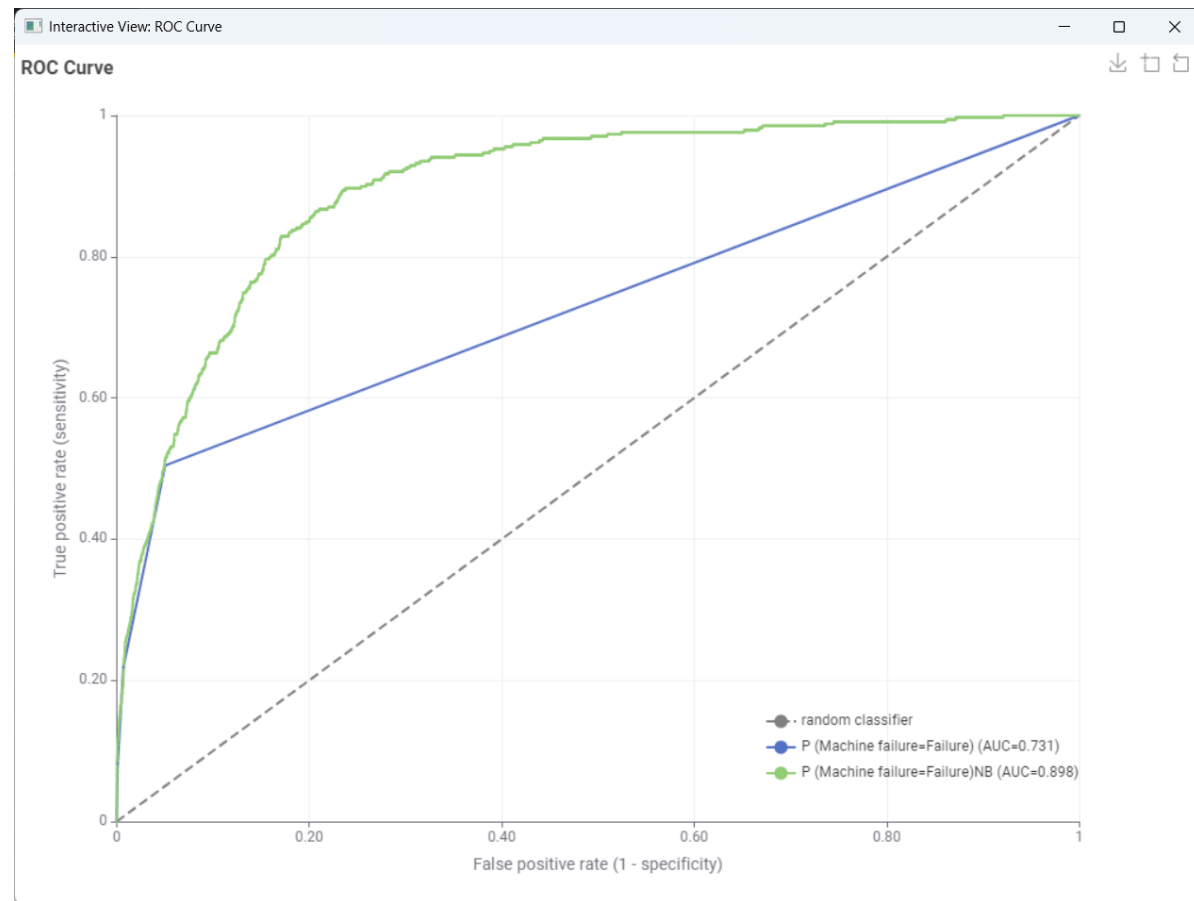
Custom suffix

Cancel Ok

Important Setting: ROC Curve



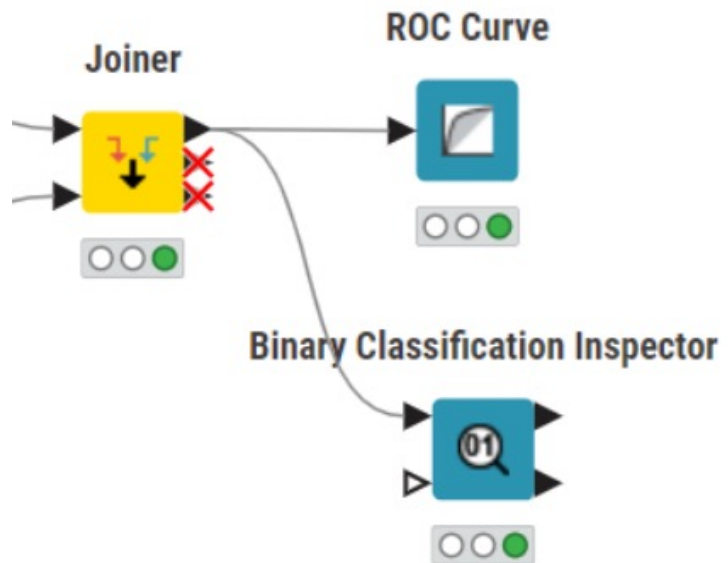
ROC Comparison



Performance Comparison: Binary Classification Inspector



KNIME: Binary Classification Inspector



Dialog - 3:30 - Binary Classification Inspector

File

Confusion Matrix Main Prediction Distribution Style Interactivity Model Stats Chart Flow Variables Job Manager Selection ROC Curve Memory Policy

Titles

Bar chart title: Scoring Metrics

Bar chart axis options

Bar chart x-axis label: Metric by Model

Bar chart y-axis label: Value

Max # tick-marks: 8

Chart options

☒ Group bars by statistics ☒ Show bar chart grid

☐ Enable bar chart panning controls ☐ Enable bar chart zoom controls

☐ Enable bar chart tooltip controls

Bar chart bars (per model)

☒ Accuracy ☐ Balanced Accuracy ☐ Error Rate

☐ False Negative Rate ☒ Precision ☒ Sensitivity

☒ Specificity ☒ F-Measure ☐ Youden's J Index

☒ AUC

OK Apply Cancel ?

Important Setting: Binary Classification Inspector

Performance: K-Nearest Neighbor



Performance: Naïve Bayes



LAB: KNIME Classification Modeling

1. สร้างแบบจำลอง Decision Tree และ Logistic Regression ด้วย KNIME
2. เปรียบเทียบ Performance ของแบบจำลองทั้ง 4 แบบจำลอง ด้วย 10-Fold Cross Validation ตามตารางต่อไปนี้

	K-Nearest Neighbor	Naïve Bayes	Decision Tree	Logistic Regression
Accuracy				
Precision				
Recall				
Sensitivity				
Specificity				
F-Score				

Q & A